‘WANT IT TO END’: TOWARDS A VIRTUAL THEATRE ENVIRONMENT

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ABSTRACT

This research and artistic work illustrates some preliminary steps taken towards creating ‘virtual theatre environments’.[1] These works are motivated by the desire to give participants an embodied, first-person experience of a specific mental state. A creative work is produced, entitled ‘Want it to End’, in which participants are led through a live performance space wearing Virtual Reality (VR) headsets. The subject material focuses on experiences of anxiety. This paper gives an overview of the approach used to create the work. A brief survey of VR and Immersive Theatre literature is used to contextualise the creative work. Using the ‘Want it to End’ example, the paper addresses aspects of the planning to realisation process of creating a virtual theatre. These aspects include ‘Metacognitive scripting’, 360-degree filming, audio composition, spatialisation and performance. In addition, curated interoceptive and exteroceptive stimulation techniques for performance are considered. Conclusions are discussed with regard to their implications for current and future project developments.

1. INTRODUCTION

1.1. Origins and Overview of Virtual Reality (VR)

French avant-garde playwright, poet and theorist Antonin Artaud asserted that illusion was not distinct from reality, advocating that spectators at a play should suspend disbelief and regard the drama on stage as reality. He is credited with the first use of the term *Virtual Reality* (Artaud & Richards, 1958).

The ‘Sensorama’, shown in Figure 1, was a machine patented in 1962 by inventor Morton Heilig. It was designed as a giant box and looked almost like a racing car simulator crossed with a photo booth (Patent No. US3050870A, 1962).



Figure 1 The “Sensorama” (Source: Engadget.com).

In 1981, French sociologist Jean Baudrillard declared that information devours its own content and that we live in a hyper-reality where we cannot distinguish reality from a simulation of reality.

Char Davie’s ground-breaking immersive virtual reality installation *Osmose* saw participants (or ‘*immersants’*, as Davies calls them) wear a Head-Mounted Display (HMD) and a body suit which monitored their breathing and balance. Depending on the rate of breathing or motion, one could find themselves embedded within a clearing, cloud or forest. Breathe too deeply and you were immersed in philosophical quotes about the clouds. Too shallow and you would find yours below the surface, observing the source code of the program (Davies, 1995).

1.2. Defining VR

Virtual Reality is not a homogenous concept. As individuals and as a society, we consume so much media that it is easy to put forward an argument that we are already engaged with copious volumes of virtual reality – through news media, video games, novels, podcasts, etc. Or that, in fact, we already live in a Virtual Reality – least a virtually *mediated* reality.

1 An intermedia theatre work, that is immersive in nature, and incorporates the use of Virtual Reality (VR) within a continuous performance.

Worthy of philosophical interrogation as this is, it does not really assist us in defining our research area and what we want to achieve through the medium termed 'Virtual Reality' or what we mean by a 'Virtual Theatre'.

Sherman & Craig (2019, p. 6-15) give five key elements essential to a Virtual Reality Experience:

1. *Participants*
2. *Creators*
3. *Virtual World*
4. *Immersion*
5. *Interactivity*

Many key elements are not solely relevant to VR. However, by outlining these parameters, Sherman & Craig reveal a unique bundle of characteristics which help to define what a VR experience is. For the purposes of this paper, we have sought to utilise Sherman & Craig's five key elements. Importantly, these elements were satisfied in the context of a virtual theatre production which utilised VR headsets.

1.3. Origins and overview of Immersive Theatre

Immersive theatre is generally regarded as theatre which breaks with notions of a fourth wall[2]. This theatre is generally not presented in an auditorium with regular seating and a stage. Often these performances invite audience participation and/or offer agency to the audience as to how they interact with the performance. This removes them from the role of spectator and repositions them as a participant.

Mandell (2016, p. 11) attempts to define clear cut attributes to the term *Immersive Theatre* asserting that, for a work to be worthy of the term, it needs to include the following:

1. A multi-sensory approach (taste, touch and smell, augmenting sight and sound);
2. A sense that the show is as much art installation as theater piece;
3. Personalization of the experience (where experiences of the audience vary, not just in interpretation of the material but in objective experience);
4. An emphasis on the social via "playful interaction";
5. The centrality of a story, no matter how fractured or obscure.

Public performance and notions of interactive performance is nothing new. Anything from Medieval theatre up to current-day street performances, could be said to satisfy the requirements of current buzz terms such as "Immersive", "Participatory" or "Site-Specific".

Nonetheless, Mandell's attributes of what constitutes an immersive theatre performance are helpful to us, if we are to work towards a broader definition of a 'Virtual Theatre'.

1.4. What is a 'Virtual Theatre'?

Defining a Virtual Theatre is difficult, and it should be noted that we are only attempting to define what it means in the context of the creation of our own artworks. We are not talking about a Virtual Theatre as a simulation for a medical operation. Nor are we talking about theatre delivered by VR, as in the theatre piece 'Staging Second Life' (Russell Fewster, Denise Wood, & Joff Chafer, 2011). This research is focused on developing a framework for creating intermedia[3] work that integrates VR with the theatre to create an unbroken, narrative-driven performance. By combining Sherman & Craig's key VR elements with Mandell's immersive theatre attributes, we can define a Virtual Theatre as:

An intermedia theatre work, that is immersive in nature, and incorporates the use of Virtual Reality within a continuous performance.

2. PLANNING FOR A VIRTUAL THEATRE EXPERIENCE

2.1. Consideration of the participant and the philosophical underpinnings of virtual theatre modality

Bay-Cheng, Kattenbelt, Lavender, & Nelson, (2010) are concerned with modes of experience within Intermedia work. They discuss five different modes of experience, which we have summarised here:

1. *Experiencer* -
   An experiencer's interaction with the performers and the technology/media creates the artistic work.

2. *Embodiment* -
   As it exists across interactions with all technology and media present in a work.

3. *Intimacy* -
   A sharing, or combination, of perceptual frames between experiencer and performer.

4. *Presence* -
   Given the nature of intermedia work, presence extends beyond spatiotemporal locality. In intermedia work, presence is negotiated using the transitional spaces between the live and the digital.

5. *Immersion* -

2 The imaginary boundary between audience and performer, generally demarcated by a stage for performers and seating for an audience.

3 Various inter-disciplinary art activities that occur between genres and disciplines (Higgins, 2001)

> The sensory experience/perception of being submerged (being present) in an electronically mediated environment. Virtual theatre can extend this to an overlap between different realities.

Bay-Cheng, Kattenbelt, Lavender, & Nelson's work can appear to be quite abstract and obtuse. As such, it is difficult to ascertain practical application for a virtual theatre context. However, the distillation of experiences into a perceptual map that the authors provide is useful in deciding what to focus on within a virtual theatre context.

2.2. Metacognitive Scripting

2.2.1. An overview

Metacognition is broadly defined as 'knowing about knowing' or 'thinking about thinking' (Flavell, 1979; Livingston, 2003; Metcalfe & Shimamura, 1994). Flavell (1979) proposed metacognition as consisting of metacognitive knowledge and metacognitive experiences. He divides this into knowledge of person variables (what we know about how we learn), task variables (what we know about the nature of a task and the processing demands it has on us) and strategy variables (what we know about strategies that we can apply to successfully accomplish a task).

It is naturally common for artists to consider the audience in the creation of any work. Indeed, the artist-audience relationship is a very large field of study in its own right (Bloom & Erlandson, 2003; Durgnat, 1970; Trivedi, 2004). It is also discussed widely in popular discourse, for instance in relation to getting a larger subscriber base on one's online media channels. However, this type of discussion is generally around the efficacy of the artist's message. When we talk about 'metacognitive scripting' we are scripting the creative material from the internal viewpoint of a participant. We are simultaneously evaluating the efficacy of our artistic message and scripting elements of their objective experience in a multisensorial manner.

2.2.2. Metacognitive scripting in the 'Want it to End' experience

The 'Want it to End' mixed reality experience started out with the following logline:

> *"Participants perform the role of a protagonist with recurring and debilitating anxiety. The world contorts and burns around them, yet they can't feel anything as a response to the situation. They experience conflicting desires of wanting it to all come to an end, whilst insisting that they stay anxious. They are haunted by their past actions as they are pointedly reminded by a strange colonial-style figure who haunts them throughout a dystopian and surreal Australian rural landscape."*

Rather than set out and script a narrative-driven story based on this logline, we scripted a set of events that we thought best conveyed the scenario. From here, we created a table which included all the staging and setting elements.

We viewed each action within the scenes as an action of the mind, rather than action within the content. We designed the instances of experience within the work with the idea that a participant may interpret them as actions, thoughts or memories. Scripting from this point allowed us to explore feelings participants might experience, and the narratives which they build around those feelings in their mind. Each 'sense-event' we created would need to facilitate the building of this internal narrative within the participant.

2.3. Filming VR content for a virtual theatre experience

2.3.1. Considering Experiencer, Embodiment, Intimacy, Presence and Immersion

Each scene was scripted from the point of view (POV) of the experiencer. The ultimate immersion of the 360-degree environment within the theatrical performance space had to be central to the planning of the filmed component. If the experiencer was to be the protagonist within our story, we had to define the characteristics of their embodiment, intimacy and presence before they were there.

Starting with the sequence of events from *Figure 2* allowed us to explore what would be going on phenomenologically and in terms of the five key modes of Experiencer, Embodiment, Intimacy, Presence and Immersion. For instance, in the campfire scene, what are possible interpretations of their surrounds is the participant having at this time? Is the participant feeling cold? Are their thoughts clear or clouded? As a result, are the sounds they hear diegetic or seemingly supernatural? Do they feel the dirt beneath them, or do they feel something else beneath them because they are visiting this space as a dream?

It was important for us to pose these questions in order to maintain an appropriate level of ambiguity. In keeping with the goal of leaving it experiential and open to interpretation, the result was a simultaneously physical and realistic world, layered over with visuals and sounds that were evocative of the supernatural. By considering each event as simultaneously a thought, memory and action, we had embedded our participants in a beautiful yet paradoxical world that made no sense except for the fact that you were undeniably inside this reality in a spatiotemporal sense. As one participant reported:

> *"I just became used to [the world] and accepted that*

skies were purple and contained fish, whilst a bushranger in a hat was giving me plastic water bottles full of strange elixirs and dogs barked in harmony with each other."

Figure 2 shows some visuals from the *Want it to End* experience. The image is a 3D image presented in 2D for the sake of publication. It features clouds and rural bushland layers.



Figure 2 Still from Desert scene of "Want it to End".

2.3.2. Creating effective Point-Of-View for enhanced embodiment

One consistent trouble in filming in 360 at the time of writing is getting a clear first person, 'embodied' experience. One often looks down in an experience to see either a black blotch (caused by rig-removal), a tripod, or simply nothing at all. Either this, or one finds themselves moving on a body but can see the head of the person that they are traveling on. Many sequences in the *Want it to End* experience required a body. They are listed here with their relevant solution:

Driving a car:

Headless mannequin with suit jacket covering the body placed in driver's seat. Custom camera attachment with a scarf wrapped around the base of the camera. Holes placed in the suit jacket. One of the authors hid themselves behind the driver's seat and placed their hands through the suit jacket and on the steering wheel. The car was then driven at a slow pace.

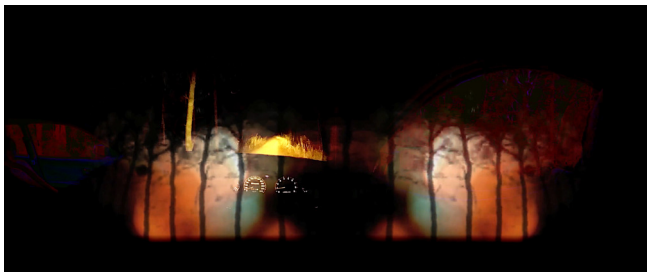


Figure 3 Night time driving scene, with flashing layers to represent potential memories of a car accident.

Walking:

One author wore the custom camera attachment with a scarf at the base. Height became a consideration in this instance and the wearer had to lower themselves so that they did not appear too tall.



Figure 4 Walking scene. Filmed in Linton Victoria.



Figure 5 Same scene as *Figure 4,* as it was experienced in the theatrical context. Performance at Aeso Studio, May 2018.

During the theatrical experience, participants were walked through a series of "sense-events" that made the kinaesthetic reality match the virtual reality they were watching in the HMDs. Figure 4 shows the vision from the HMD perspective. Figure 5 shows the theatrical performance. Curated smell sensations are occurring whilst the participants are led barefoot over grass.

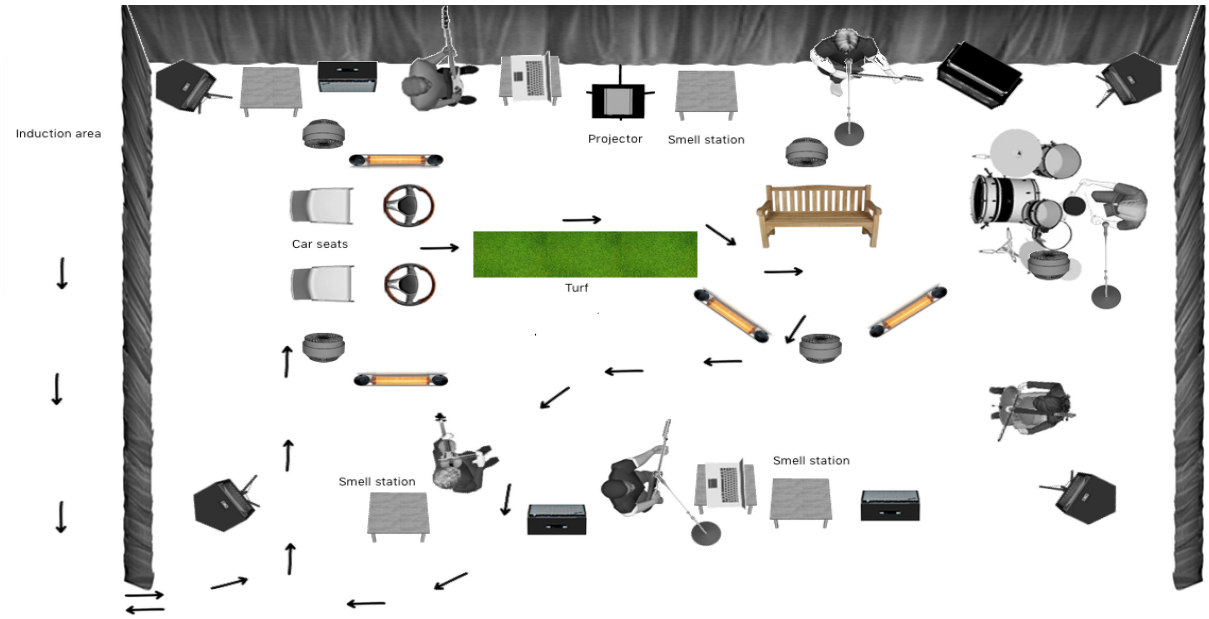


Figure 6 "Want it to End" stage plot. Includes interoceptive and exteroceptive tools, performers, technology, direction of audience flow, relevant props.

2.4. Audio and composition for a virtual theatre environment

2.4.1. Considering Experiencer, Embodiment, Intimacy, Presence and Immersion

In a virtual theatre context, involving the use of HMDs for the period of time that the participant is within the headset, the participant exists in a different visual reality to the performers. As composers, sound designers and performers, we can start by considering ourselves as something of a 'universe-jumper'. Performers of the music are physically co-present to participants who are mentally with the performance. As such, musicians are 'performing on' their audiences, rather than 'performing to' them.

2.4.2. *Diegetic levels of sound within a virtual theatre experience*

According to Gorbman (1976), sound source on the narrative level may be diegetic[4], extra-diegetic[5], and meta-diegetic[6]. Meta-diegetic is sound imagined, or perhaps, hallucinated by a character. In the 'Want it to End' experienced, we started with the premise that all sounds made in the space could represent any level of diegesis. Given our premise that all stimuli could be interpreted as thoughts, memories or actions, it seemed logical to imagine that all three were at play within the musical composition. Diegetic sounds that corresponded with the visual sounds (e.g. walking) were played and embedded into the music. Meta-diegetic sounds were to correspond with much of our metacognitive scripting. For instance, if the participant was scripted to feel intense anxiety at a specific point, dissonant strings could be used. In terms of the extra-diegetic, we utilised Fierce Mild's song 'Lost Highway', which was also performed live. This song was broken up with additional scoring and performance. We acknowledge that – being that participants were characters themselves – they obviously heard this music. However, its function was very similar to the extra-diegetic sound of a film as gave cues to the participants as to how to feel about the imagery.

2.4.3. Spatialisation

Complex sound fields have been integral to 'immersive' art environments since at least the 1920s (Wendel, 2012). So too is it integral to a virtual theatre space. The ultimate goal would be something of a high order ambisonic speaker array combined with different elevated positions and spatialisation of all instrumental players and singers. However, in the 'Want it to End' experience we were limited to forty square meters of space. Within this space, at one time, we had 5 musical performers with all of their equipment, 2 guides and 2 participants who roamed the space. With a quadraphonic PA set up, spaced out amplifiers and a large drum kit, performance and timbre became key to best immersing the participant in the sound world.

2.4.4. Instrumentation and timbre selection

The 'Want it to End' experiences contained the following instrumentation:

Fierce Mild ensemble

- Voice
- Electric guitar (prepared and unprepared)

4 Sound experienced by the characters
5 Sound not experience by the character, but heard by the audience

- Bass guitar
- Cello
- Violin
- Drums

Additional Instrumentation

- Percussion – Djembe, seedpod shaker, Tambourine processed through effects
- Jaw-Harp
- Hapi Drum
- Samples, Ableton effects, Sound Toys, Altiverb, Valhalla, Occulus Spatialiser, Envelop for Live

2.5. Performing audio in 'Want it to End'

As musical performers we were 'performing to' our audience within a singular location. However, two subjective realities existed. Each interaction is innately different. One begins to sense the disposition and relative levels of comfort of participants. Some were petrified, whilst others were open to anything that came their way. The seriousness of 'performing' was less as the audience could not view us. This gave us scope for clearly communicating with each other about a sound event (such as a gong sound behind the participant's head) without giving anything away about what we were planning.

Sometimes we would whisper in the participant's ears or breathe slightly on their neck. Given the aesthetic merging of electronic soundscapes with rock, it became pivotal to know when to move from the channel strip[7] to direct interaction with a participant. A scream would first appear within amplified, quadrophonic sound, then it would fade, only to back right behind the participants ear. The sound transitioned from a distanced full-bodied performance, to an intimate encirclement within the space of 10-30 seconds.

2.6. Curating sensation to create embodiment

By definition, our virtual theatre has to be multi-sensory in nature. We are interested here in exploring the ability of sensory stimulation within performance to create a deeper sense of embodied experience which could perhaps cultivate societal empathy towards neuro-diverse people – such as those experiencing anxiety disorders. We can do this by examining the results of stimulating different aspects of the somatosensory[8] system throughout a virtual theatre performance.

Klich (2019) refers to a 'visceral dramaturgy' in her practice-as-research project *Daphne in three Movements*. She argues that employing a visceral dramaturgy can facilitate a heightened awareness of one's own proprioception[9], viscerception[10], and sense of active being and functioning in the world. It is unclear whether her work solely focuses on the viscera or whether this a term used to describe an approach to theatre that attempts to elicit more interoceptive[11] awareness in the audience. However, it is a useful term when describing a type of theatre which seeks to direct an audience's attention towards their inner mental and emotional states.

Indeed, much art has been concerned with the viscera and interoceptive responses. Chaim Soutine painted images of dead rabbits and slabs of beef in thick oil paintings. When viewing these, one may feel or imagine the sensation of particular viscera stretching and contorting. Seth (2019), using the prism of Expressionist art, argues that visual experience is shaped by interoceptive predictions, and in turn emotional responses. Therefore, whether consciously or not, it is integral to any art-making process.



Figure 7 Chaim Soutine's *Fish, Pepper Onions* (c. 1919)

2.6.1. A theatre of multiple senses – the Want it to End example

The 'Want it to End' experience made use of the following senses: smell, taste, thermoception (heating and cooling; wind direction), hygroreception (detection of moisture content) somatosensory (touch), proprioception and misattribution.

Verbal testimonies from participants during informal conversations after the experience have led us to suspect that these senses created a much deeper sense of embodiment to the experience, and a deeper sense of engagement with the work, than if they had not been included. Further, more empirical, research would be required to state this conclusively.

2.7. Scoring for a multisensory composition

In terms of a score, musicians followed a cue-sheet which was made using the same video but with sense and emotion event cues.

7 Meaning to move from performing with amplified sound through a mic to acoustically projecting the sound into the space
8 The system in the human body, of sensory neurons and pathways, that responds to changes at the surface or inside the body
9 Sense of body position and movement
10 Senses arising from stimulation of the viscera (internal organs)
11 The sense of the internal state of the body

3. TECHNICAL CONSIDERATIONS

At the time of research, VR is still considered an emerging medium and there are many 'bugs' associated with working in this environment – particularly works that involve HMDs. Seemingly simple operations can quickly turn into logistical nightmares when a technology is young and without a 'black box' model.[12]

This section gives a basic overview of some of the challenges faced and discusses some resolutions. Whilst this research will do it's best to explain the technical issues at play, it is written from the perspective of an artist and not a technology expert.

3.1. Streaming VR content to multiple devices

Multicasting to headsets presents significant challenges with current technologies. In researching this area, we found many 'turn-key solutions'[13] on offer from private companies and individuals who closely guarded their information to limit competition and safeguard profits. Given resource constraints, we had to embark on the creation of our own set of solutions to the technological issues at play. Based on our findings, we have listed some considerations for integrating VR headsets into a multimedia project of any kind.

3.1.1. Streaming and Syncing files

A basic understanding of computers and networking comes in handy when dealing with a virtual theatre arrangement. Put simply there is a 'server-side', which is controlled by a computer; and a 'client-side' which is the HMD.

For the 'Want it to End' project, we streamed the video content from a Windows laptop, using Skybox VR player to two HMDs (Samsung Gear VR). We first attempted this using a closed network on a TPLink 300mbps router. However, there were issues with port-blocking which meant that the server and client could not detect each other. As such, we chose a work-around that made use of a tethered internet connection from an iPad. This situation was not ideal but did work. Our 'score for the senses' (Section 2.7) was sent over a different signal to its own projector.

3.1.2. File size and quality issues

The current quality of 360-degree filming and playback equipment already poses a challenge. The file sizes are large, and this can create a lot of latency across a network streaming the content. Compressing the files in the 'Want it to End' experience became unfortunately necessary – from approx. 8GB in size, to 1GB.

Part of the bottle neck exists with processing power too. At present, the race is on to develop 'foveated rendering' - a graphics-rendering technique that reduces the processing requirements by greatly reducing the image quality in the peripheral vision within a headset – allowing a processor to focus on where a participant is looking at a given point in time.

3.1.3. Proprietary tools and the need for preloaded content

Some companies offer 'turn-key solutions' to the multiple headset problem. A company called VR SYNC offers monthly licenses at a considerable cost. AV hire companies also rent out this system or similar at high costs.

3.1.4. Developing own interface using Unity 3D

For our future work – *'The Door in Question'* we are developing our own closed system through the software Unity 3D. This will enable us to preload content onto headsets and trigger it from our server-side without having to stream large volumes of data over a network.

3.2. HMDs ability to disrupt the flow of experience in a hybridised virtual theatre

A final but important consideration is the imposing and 'clunky' nature of using a VR headset or HMD. First performed in 2017, Frogman is a production by *Curious Directive* which takes the form of a hybridised virtual theatre. The story centres on a missing girl in Queensland, Australia. The disappearance occurred in the mid 1995. The physical element of the play is set in present day, whilst the VR component is set in 1995. The audience take the role of jurors, who are instructed to put on their headsets at given moments by the actors to 'assess evidence' that is presented through the headset.

Many reviews gave a fairly dim assessment of the play and complained that the narrative was bogged down by gimmickry, for example Woodhead, 2018. The 'Want it to End' experience only had a singular transition moment. However, our future work focuses on exploring continual transitions in a seamless way. As such, scripting for this 'clunkiness' is no doubt an imperative.

Much of what worked for us in *Want it to End* was an abstract and surreal aesthetic which blurred the lines of reality rather than attempting to replicate or simulate it – which much VR does. Though beyond the scope of our discussion here, it may well prove that the medium of VR (in headset form) is more suited to abstract and surreal works. Either way, it is important to consider the character that this new medium can impose on any production and to let the overarching experience determine the ultimate technology necessary to convey the artistic intent.

12 A model whereby the user inputs a command and the computer system produces a result, requiring little to no knowledge on the user's part of how the system works.

13 A type of system built end-to-end for a customer that can be easily implemented into a current business process.

4. REFLECTIONS

4.1. Evaluating the 'Want it to End' experience

The 'Want it to End' experience was initially intended as a way of exploring new frontiers in the way music concerts are presented. As an audio-visual band, Fierce Mild has always considered this a natural evolution. 'Want it to End' was relatively successful. Tickets to the initial, self-produced iteration sold out. The experience was then booked by several local municipalities as part of wider festival umbrellas – West Projections Festival and Melbourne Music Week. This gives us reason to assert that there is a commercial viability in this type of hybridised virtual theatre work; that patrons are indeed very interested in deeper embodiment within their interactions with artistic works.

Through qualitative analysis in the form of participant testimonials, it appeared evident to us that there exists an ability to create deeply embodied via the use of interoceptive and exteroceptive device. Some audience members were so deeply affected by the experience that they requested some time to process what they had experienced before returning to comment on the work to us. It is not possible to say, from our limited testimonials alone, whether or not participants experienced an embodied state of anxiety. Further research that is more empirically motivated is required to assess the efficacy of this type of experience in providing such embodiment.

The terminology around VR is in a state of flux. Many people consider it to be very different things. Indeed, it was difficult to convey through the marketing of the 'Want it to End' experience, exactly what it was. There is often much debate about the success and failures of the technology and whether 'VR is the future' or 'dead in the water'. For us, this debate is frivolous. Artists will utilise whatever tools are around them at a given point in time. Bearing these debates in mind though, it is important to stress that HMDs in the context of the creation of a Virtual Theatre are replaceable with other types of technology. Any reader of this work may feel free to consider the research with any tools in mind that they see fit to use.

4.2. Future work – 'The Door in Question'

'*The door in Question*' is a virtual theatre work exploring an experience of Schizophrenia. Many scenes are created using the writings of the artist's mother, who was diagnosed with schizophrenia. It is also inspired by the artist's lived experience with psychosis.

The creative output focuses on heightened states of paranoia and mental disorganisation. The materials are presented to encourage audiences to gain a sense of understanding to the way that the emotions of love, anger and sadness are expressed within these states. An intermedia work is created which explores methods of utilising exteroceptive and interoceptive techniques as well as visceral dramaturgies in order to give effect to a schizophrenic experience. This work focuses on the faultline between virtuality and physical reality as a worthy point of exploring an 'illness narrative' in a theatrical context. The purpose is to create an artwork utilising lived experience and qualitative research in order to give an engaging representation of the experience that is also truer to life. As such, the research aims to encourage a deeper sense of empathy and understanding towards people who experience schizophrenia and psychosis amongst the community.

This work aims to create several successful transitions between the virtual and physical worlds, exploring the Faultline between the real and the virtual. The techniques developed from the 'Want it to End' experience will be further developed, refined and expanded upon with a similar goal of creating a continuous multi-modal technological based performance which encourages a deeper sense of embodiment.

5. REFERENCES

Bay-Cheng, S., Kattenbelt, M. J., Lavender, A., & Nelson, R. (2010). *Mapping Intermediality in Performance*. Amsterdam University Press.

Cuisick, W. (2016, February). *Virtual Reality and Theater: Simulacra and Simulation*. Ted Talk. Retrieved from https://www.youtube.com/watch?v=SEgSkfflgvg

Durgnat, R. (1970). ART AND AUDIENCE. *The British Journal of Aesthetics*, *10*(1), 11–24. https://doi.org/10.1093/bjaesthetics/10.1.11

Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive–developmental inquiry. *American Psychologist*, *34*(10), 906–911. https://doi.org/10.1037/0003-066X.34.10.906

Gorbman, C. (1976). Teaching the soundtrack. *Quarterly Review of Film Studies*, *1*(4), 446–452. https://doi.org/10.1080/10509207609360969

Higgins, D. (2001). Intermedia. *Leonardo*, *34*(1), 49–54. Retrieved from JSTOR.

Klich, R. (2019). Visceral Dramaturgies: Curating Sensation in Immersive Art. Body, Space & Technology, 18, 175–197. https://doi.org/10.16995/bst.319

Kostov, G. (2015). *Fostering Player Collaboration Within a Multimodal Co-Located Game*. https://doi.org/10.13140/RG.2.1.3865.7044

Livingston, J. (2003). *Metacognition: An Overview*.

Mandell, J. (2016, February 9). Immersive Theatre, Defined. Retrieved from https://howlround.com/immersive-theatre-defined

Matthews, D. (2018, January 16). Amount of pixels needed to make VR less crap may set your PC on fire. *The Register*. Retrieved from

https://www.theregister.co.uk/2018/01/16/human_limits_of_vr_and_ar/

Merleau-Ponty, M. (1962). *Phenomenology of Perception*. The Humanities Press.

Metcalfe, J., & Shimamura, A. P. (Eds.). (1994). Metacognition: Knowing about knowing. *Metacognition: Knowing about Knowing*., xiii, 334–xiii, 334.

Piumsomboon, T., Day, A., Ens, B., Lee, Y., Lee, G., & Billinghurst, M. (2017, November 27). Exploring enhancements for remote mixed reality collaboration. Presented at the SIGGRAPH Asia 2017 Mobile Graphics & Interactive Applications. https://doi.org/10.1145/3132787.3139200

Russell Fewster, Denise Wood, & Joff Chafer. (2011). Staging Second Life in Real and Virtual Spaces. In Giovanni Vincenti & James Braman (Eds.), *Teaching through Multi-User Virtual Environments: Applying Dynamic Elements to the Modern Classroom* (pp. 217–235). https://doi.org/10.4018/978-1-61692-822-3.ch013

Seth, A. K. (2019). From Unconscious Inference to the Beholder's Share: Predictive Perception and Human Experience. *European Review*, 1–33. https://doi.org/10.1017/S1062798719000061

Sherman, W. R., & Craig, A. B. (2019). Chapter 2—VR: The Medium. In W. R. Sherman & A. B. Craig (Eds.), Understanding Virtual Reality (Second Edition) (pp. 60–100). https://doi.org/10.1016/B978-0-12-800965-9.00002-7

Soutine, C. (1919). Fish, Pepper, Onions [Painting].

Steuer, J. (1992). Defining Virtual Reality: Dimensions Determining Telepresence. *Journal of Communication*, *42*(4), 73–93. https://doi.org/10.1111/j.1460-2466.1992.tb00812.x

Thurley, K., & Ayaz, A. (2016). Virtual reality systems for rodents. Current Zoology, 63(1), 109–119. https://doi.org/10.1093/cz/zow070

Wendel, D. D. B. (2012). The 1922 'Symphony of Sirens' in Baku, Azerbaijan. *Journal of Urban Design*, *17*(4), 549–572. https://doi.org/10.1080/13574809.2012.706366

Woodhead, C. (2018, October 12). 'First virtual reality play' bogged down by gimmickry. *The Sydney Morning Herald*. Retrieved from https://www.smh.com.au/entertainment/theatre/first-virtual-reality-play-bogged-down-by-gimmickry-20181012-p509c8.html

Zangwill, N. (1999). Art and Audience. *The Journal of Aesthetics and Art Criticism*, *57*(3), 315–332. https://doi.org/10.2307/432197